

# Centre of Gravity

## Balancing Acts

Discover the magic balancing point  
with everyday Indian objects!



# Meet the Dancing Doll of Tanjore

This doll never falls down — can you  
guess why?

Let's find out the secret of its  
magic balance!





# The Magic Balancing Point

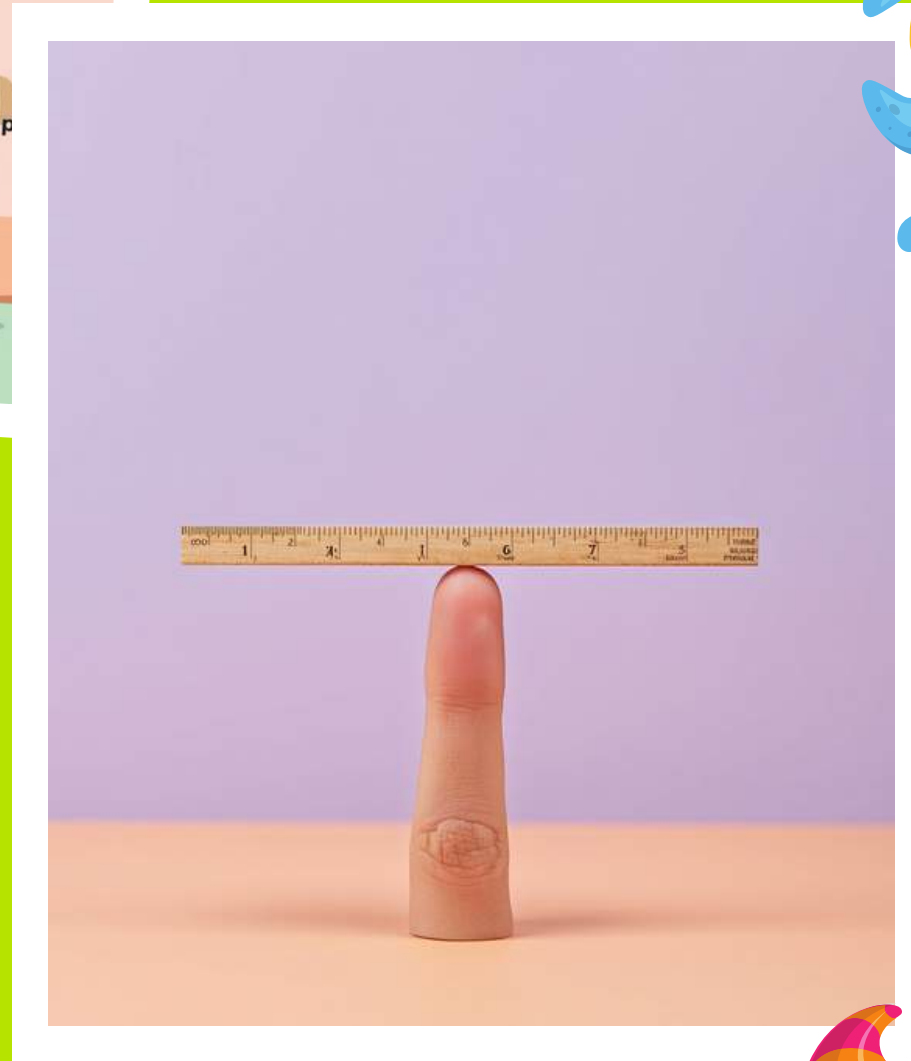
The Centre of Gravity is your object's magic balancing spot!

- If you hold an object exactly at this spot, it won't fall.
- This spot depends on how the weight is spread out.



# Finding the Balancing Point

The bat's heavy handle pulls the magic spot closer to one end. The scale balances right in the center because weight is spread evenly. Look for the glowing green "Gogo" dot — that's where each object's magic balancing point is!







# Why Does the Heavy Bottom Matter?

Objects with a low magic spot don't fall easily! Heavy bottoms keep things steady and hard to tip over.

# Changing Our Body's Centre of Gravity

When we move our arms or legs, our magic balancing spot moves too! This is why dancers practice special poses to stay balanced.

Wide, low stances help dancers stay steady during fast spins and moves. The lower your magic spot, the harder it is to fall!



# The Stand-Up Challenge

Try this fun activity to feel how balance works!

- Sit straight on a chair, feet flat on the floor, arms crossed.
- Now try to stand up without leaning forward.
- It's tricky because your magic spot stays over the chair!





# ● Building a Steady Tower

## Steel Dabbas

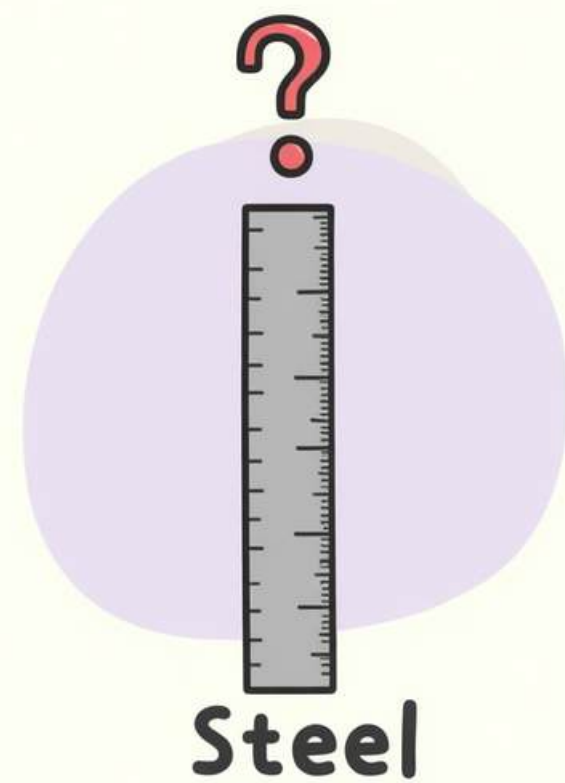
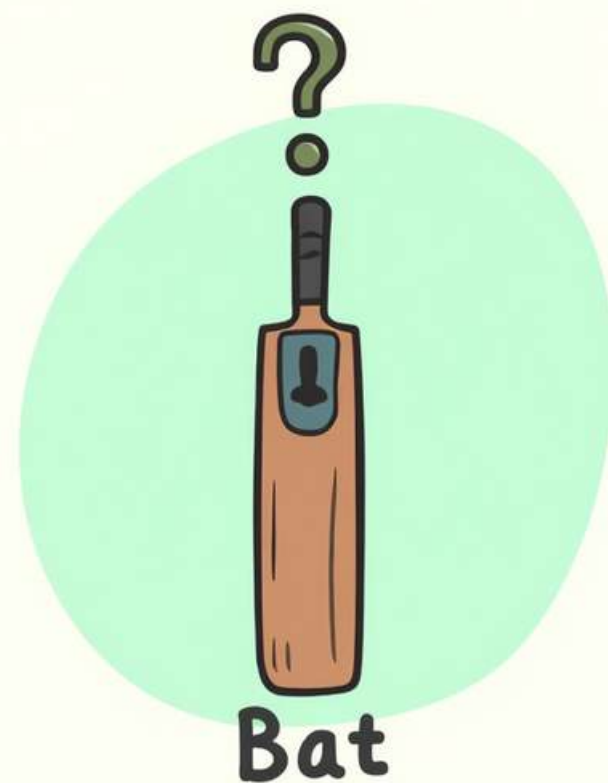
**Big base + low magic spot = super steady tower!**  
**Try this at home with your lunchboxes!**  
**Stack them biggest at bottom for stability.**





# Quick Quiz:

## Where's the Magic Spot?



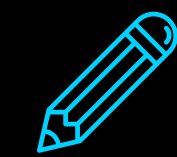
### Cricket Bat

Where does it balance? Closer to the handle or in the middle? The heavy handle pulls the magic spot!



### Steel Ruler

Can you find the balancing point? It's exactly in the middle because the weight is spread evenly!



### Round Thali

Where's the magic spot on a circular plate? Right in the center — Gogo shows you!





Quick Quiz:

Which Posture Is More Stable?

Who will stay steadier during a fast turn? The wide stance wins! A lower, wider position keeps your magic balancing spot closer to the ground.



# What Did We Learn Today?



## **Magic Balancing Spot**

The Centre of Gravity is your object's special point where it balances perfectly!



## **Heavy Bottoms Help**

Objects with heavy bottoms are hard to tip over — they stay steady and stable!



## **Balance Changes**

Your body's balance shifts when you move your arms and legs. Find balance everywhere!

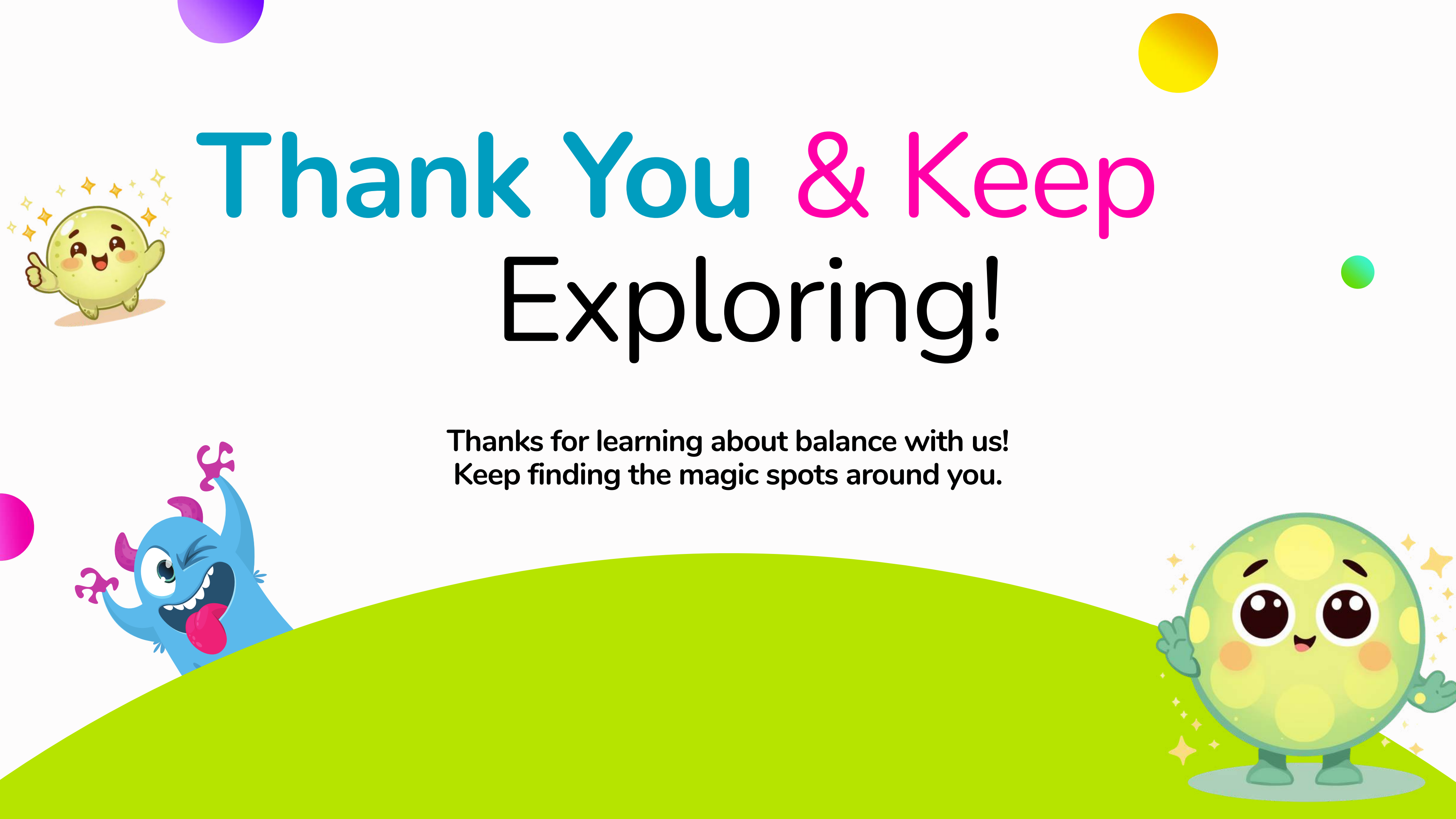


# Your Turn — Try This at Home!

- Balance a steel ruler or cricket bat on your finger — find the magic spot!
- Stack your lunchboxes with the biggest at the bottom for a steady tower.
- Try the stand-up challenge with a family member and see who can do it!







# Thank You & Keep Exploring!

Thanks for learning about balance with us!  
Keep finding the magic spots around you.